DAY # 8 [16/9/25]

Power Cables :

Materials :

Core / Conductors - Al, Cu

Insulation - Rubber, PVC, XLPE (cross linked polyethylene)

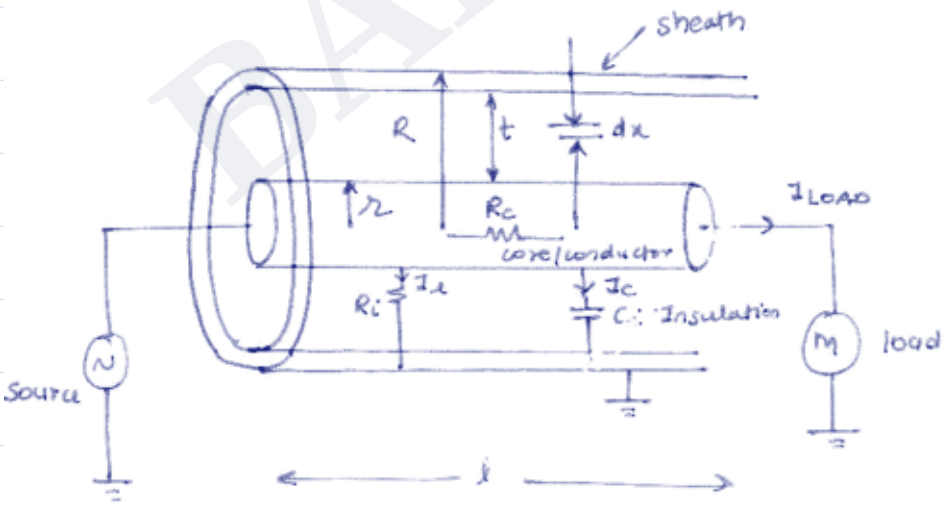
Sheath / Armour - Lead, steel, Galvanised iron.

Parameters of cable :

(1) Cable (or) Insulation Resistance (R_i)

(2) Conductor (or) Core Resistance (R_c)

(3) Cable (or) Insulation Capacitance (C_i)



Cable cross-section: core radius r, overall radius R, insulation thickness t, sheath; core resistance Rc, insulation resistance Ri with leakage current Il, capacitance Ci with charging current Ic; length l; source and load

r = Radius of core ; R = overall radius of cable ; t = Thickness of insulation = R - r

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I_l = leakage current ; I_c = charging current

Parameters of Cable :

(1) Insulation Resistance (R_i)

$$dR_i = \frac{\rho_i (dx)}{2\pi x l} ; \quad R_i = \frac{\rho_i}{2\pi l}\int_r^R \frac{dx}{x}$$

$$\boxed{R_i = \frac{\rho_i}{2\pi l} \ln\left(\frac{R}{r}\right)} \quad \backslash Omega/m$$

ρ_i = Resistivity of insulation ; $R_i \propto \frac{1}{l}$; $\frac{R_2}{R_1} = \frac{l_1}{l_2}$

(2) Core Resistance (R_c)

$$R_c = \frac{\rho_c l}{\pi r^2}$$

$R_c \propto l$; $\frac{R_{c2}}{R_{c1}} = \frac{l_2}{l_1}$; ρ_c = Resistivity of core

(3) Cable Capacitance : q = line charge

$$E = \frac{q}{2\pi\epsilon_o\epsilon_r x} ; \quad V = \int -E\,dx = -\int_R^r \frac{q}{2\pi\epsilon_o\epsilon_r x} dx$$

$$\boxed{V = \frac{q}{2\pi\epsilon_o\epsilon_r} \ln\left(\frac{R}{r}\right)}$$

$$\boxed{C_i = \frac{q}{V} = \frac{2\pi\epsilon_o\epsilon_r}{\ln\left(\frac{R}{r}\right)}} \quad F/m$$

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$$\boxed{I_c = 2\pi f C_i V_i} \quad A/m$$

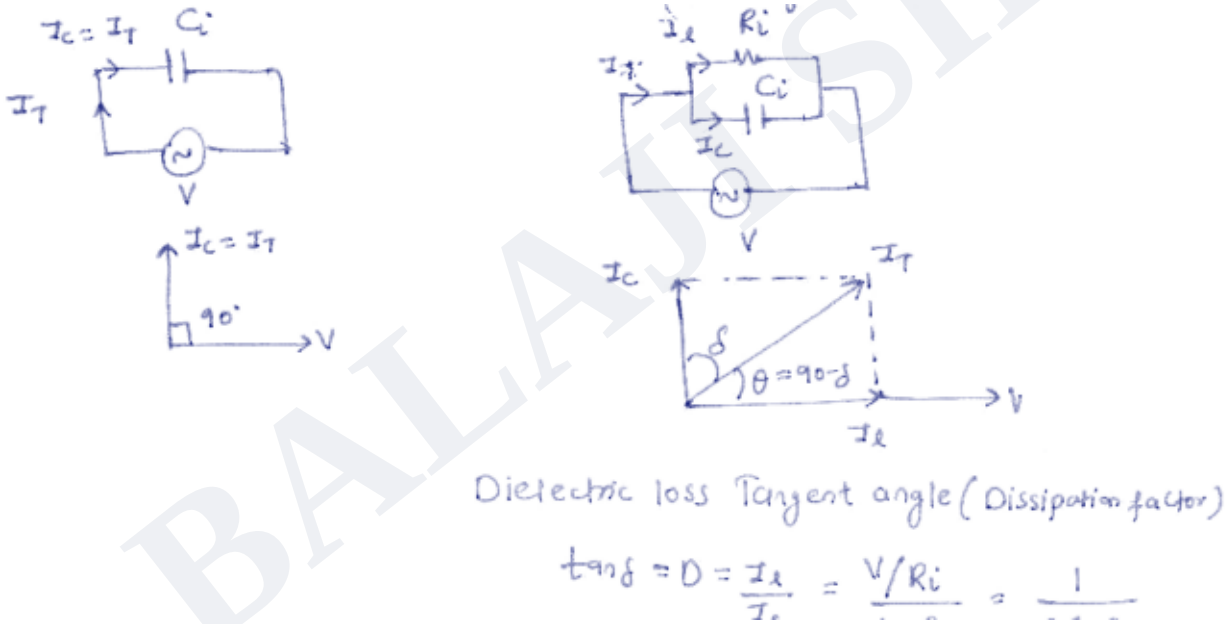
Eg: R_i = 495 MΩ/km ; core diameter = 2.5 cm ; $\rho_i = 4.5 \times 10^{14}$; t = R - r ; l = 1 km = 1000 x 100 cm

$$R_i = \frac{\rho_i}{2\pi l} \ln\left(\frac{R}{r}\right) \Rightarrow 495 \times 10^6 = \frac{4.5 \times 10^{14}}{2\pi (1000 \times 100)} \ln\left(\frac{R}{2.5/2}\right)$$

R = 2.5 cm ; t = R - r = 2.5 - (2.5/2) = 1.25 cm.

Ideal power loss of cable :

Ideal cable [$R_i = \infty$] and # Practical cable [$R_i \neq \infty$]



Dielectric loss Tangent angle (Dissipation factor)

$$\tan\delta = D = \frac{I_l}{I_c} = \frac{V/R_i}{\quad} = \frac{1}{\quad}$$

Ideal cable: only Ci, Ic = IT, 90 degrees ahead of V. Practical cable: Ri in parallel with Ci, leakage Il and charging Ic; phasor diagram with IT, angle delta, theta = 90 - delta

Dielectric loss Tangent angle (Dissipation factor) :

$$\tan\delta = D = \frac{I_l}{I_c} = \frac{V/R_i}{V\omega C_i} = \frac{1}{\omega C_i R_i}$$

Dielectric power Loss (P_l) :

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$$P_L = \frac{V^2}{R_i} = V^2(\omega C_i)\tan\delta \quad W/ph$$

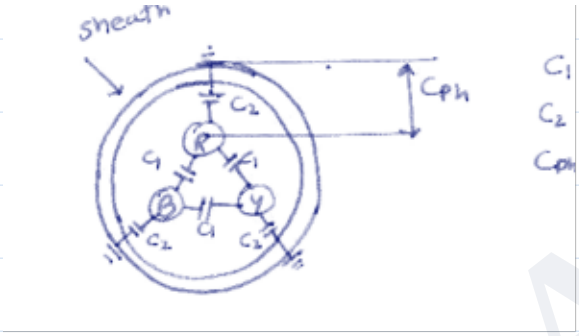
$V = V_{ph} = \frac{V_L}{\sqrt{3}}$; p.f. of cable = $\cos\theta = \cos(90-\delta) = \sin\delta$

Eg: 110 kV, f = 50 Hz, C = 125 nF/km, Dielectric loss tangent of XLPE = 2×10^{-4}.

Dielectric power loss :

$$P_L = V^2(\omega C_i)\tan\delta = \left(\frac{110\times10^3}{\sqrt{3}}\right)^2 (2\pi \times 50) \times 125 \times 10^{-9} \times 2 \times 10^{-4} \approx 317$$

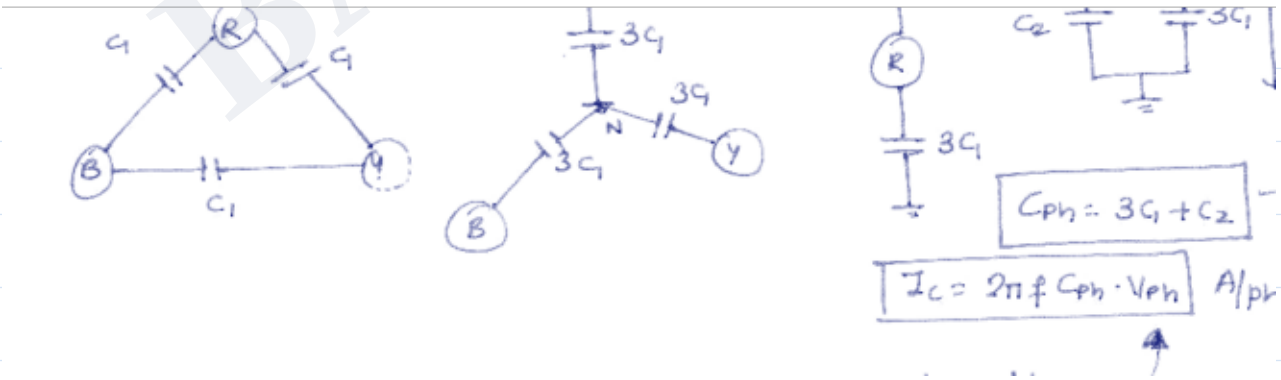
Capacitance per phase of a 3-ϕ or 3 core cable :



Three-core cable inside sheath: C1 between phases, C2 between phase and sheath, Cph capacitance per phase

C_1 = ph-ph capacitance ; C_2 = ph-sheath capacitance ; C_{ph} = Capacitance per phase.

Convert Δ to Y : each C_1 of the delta becomes $3C_1$ to neutral N



Delta of C1 converted to star of 3C1; per phase C2 in parallel with 3C1 gives Cph

$$C_{ph} = 3C_1 + C_2$$

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$$I_c = 2\pi f C_{ph} \cdot V_{ph} \quad A/ph$$

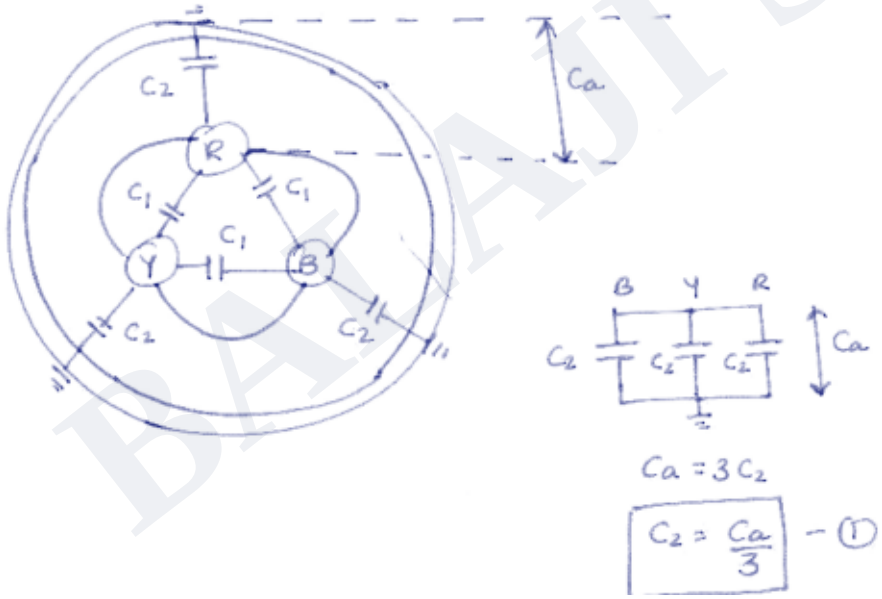
$$V_{ph} = \frac{V_L}{\sqrt{3}}$$

Practical Measurement of C_1, C_2 and C_{ph} :

(1) Connect all the conductors R, Y, B together; measure the capacitance from one of the phase to the sheath; from this calculate capacitance C_2.

(2) Connect any of the two conductors to the sheath and measure the capacitance between remaining 3rd conductor to the sheath; from this calculate capacitance C_1 and C_{ph}.

Step 1 :



All three conductors R, Y, B joined: measured capacitance Ca = three C2 in parallel

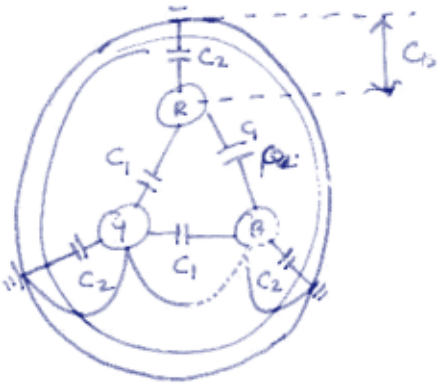
$$C_a = 3C_2$$

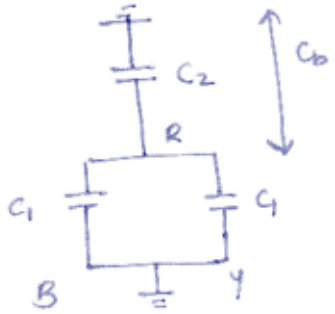
$$C_2 = \frac{C_a}{3} \quad -(1)$$

Step 2 :

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P (2):

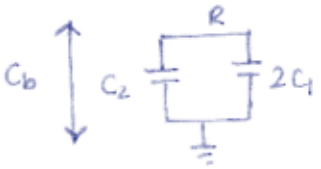


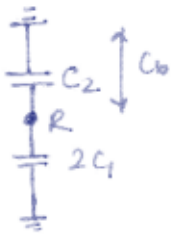


$C_b = 2C_1 + C_2$

$= 2C_1 + \frac{C_a}{3}$

$C_1 = \frac{3C_b - C_a}{6}$ — (2)





Two conductors Y, B connected to sheath; measured between R and sheath: Cb = C2 in parallel with 2C1

$$C_b = 2C_1 + C_2 = 2C_1 + \frac{C_a}{3}$$

$$C_1 = \frac{3C_b - C_a}{6} \quad -(2)$$

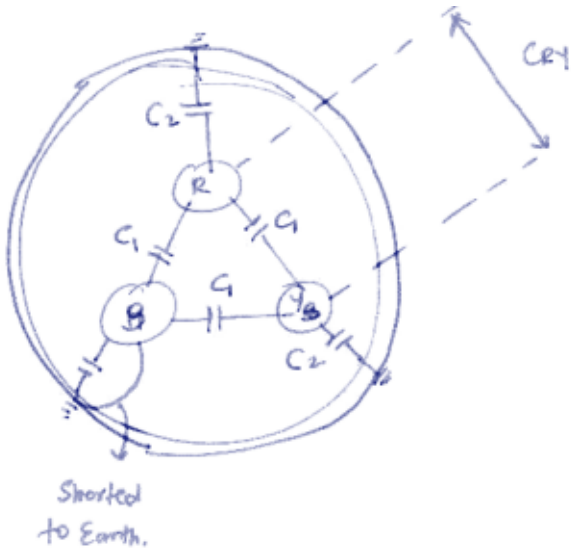
Substitute C_1 and C_2 in :

$$C_{ph} = 3C_1 + C_2 = 3\left(\frac{3C_b - C_a}{6}\right) + \frac{C_a}{3}$$

$$C_{ph} = \frac{9C_b - C_a}{6}$$

Special case :

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$$C_{RY} = \frac{(3C_1 + C_2)}{2} = \frac{C_{ph}}{2}$$

$$C_{ph} = 2(C_{RY})$$

Three-core cable with one conductor shorted to the sheath (earth); capacitance C_RY measured between the other two cores

$$C_{RY} = \frac{(3C_1 + C_2)}{2} = \frac{C_{ph}}{2}$$

$$C_{ph} = 2(C_{RY})$$

Eg: The kVA taken by 10 km long 3-ϕ 3-core cable. If the capacitance measured between any two cores with third core shorted is 0.3 $\mu F/km$, when it is connected to 10 kV, 50 Hz bus bar. (a) 142.6

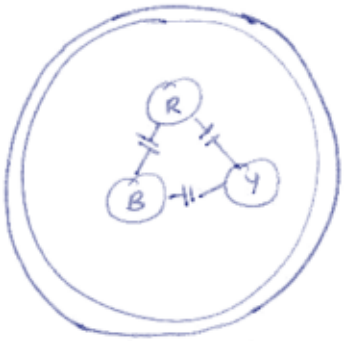
$$C_{ph} = 2(0.3) = 0.6$$

$$Q_{c3\phi} = 3(2\pi f C_{ph} V_{ph}^2) \,/ \, @ \; V_{ph} = \frac{V_l}{\sqrt{3}}$$

$$Q_{c3\phi} = 3\left(2 \times \pi \times 50 \times 0.6 \times 10^{-6} \times \left(\frac{10 \times 10^3}{\sqrt{3}}\right)^2\right) \times 10 = 188.5 \text{ kVAR}$$

Eg:

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Three-core cable R, Y, B with capacitances between cores

$$I_c = 2\pi f C_{ph} V_{ph} = 2\pi f(3C_1 + C_2)\frac{V_l}{\sqrt{3}} = 2\pi \times 50(3 \times 0.2 + 0.4) \times 10^{-6}\frac{11 \times 10^3}{\sqrt{3}} \approx 2\,A$$

Electric field stress / Vtg Gradient of cable.

* In case of power cables, electric field is maximum for the insulating layer which is close to the core or conductor and minimum for the insulating layer which is close to the sheath.
* This difference of electric field produces vtg gradient which will increase the leakage current so that heating is increased and performance of cable is affected.

$$E = \frac{q}{2\pi\epsilon_0\epsilon_r x} \quad -(1)$$

$$V = \frac{q}{2\pi\epsilon_0\epsilon_r}\ln\left(\frac{R}{r}\right) \Rightarrow \frac{q}{2\pi\epsilon_0\epsilon_r} = \frac{V}{\ln(R/r)} \quad -(2)$$

Substitute (2) in (1):

$$\boxed{E = \frac{V}{x\ln(R/r)}}$$

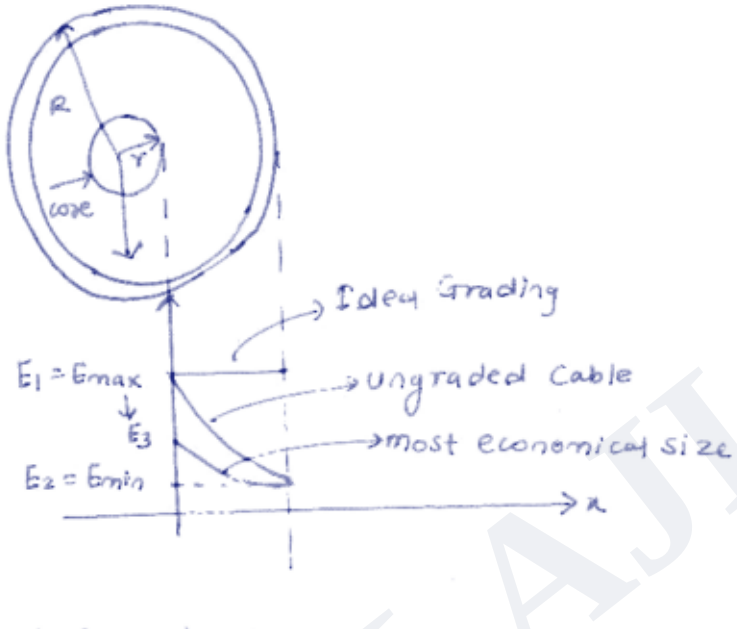
(1) At x = r = surface of core:

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$$E_{max} = \frac{V}{r \ln\left(\frac{R}{r}\right)}$$

(2) At x = R = nearer to sheath :

$$E_{min} = \frac{V}{R \ln\left(\frac{R}{r}\right)}$$



Cable cross-section with core radius r and overall radius R; plot of E against x: ungraded cable falls from E1 = Emax to E2 = Emin, ideal grading is a flat line, most economical size gives E3 in between

Methods of Improving performance of cable

(1) Most Economic size of cable

(2) Grading of cable - method of maintaining uniform electrical field through thickness of insulation.

(a) Capacitance Grading

(b) Inner / Inter sheath Grading

(1) Most Economic size of cable

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$$E = \frac{V}{r\ln\left(\frac{R}{r}\right)}; \quad \text{to minimize } E: \frac{d}{dr}\left[r\ln\left(\frac{R}{r}\right)\right] = 0$$

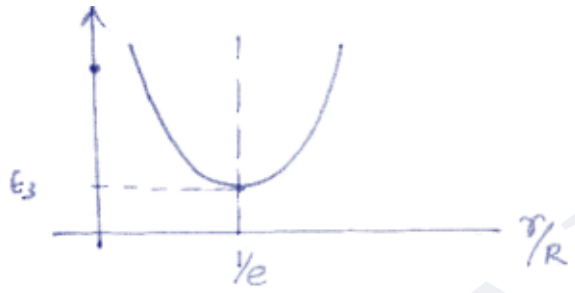
$\frac{R}{r} = e = 2.718$

t = Thickness of insulation = $R - r = 2.718r - r$

$$\boxed{t = 1.718r}$$

At $\frac{R}{r} = e$:

$$\boxed{E = \frac{V}{r\ln(e)} = \frac{V}{r} = E_3}$$



Plot of E against r/R with minimum E3 at r/R = 1/e

Eg: Determine the economic overall diameter of single core cable metal sheathed for a working vtg of 75 kV. If the dielectric strength of insulating material is 60 kV/cm.

-> peak vtg of cable = 75 kV ; E_{max} = 60 kV/cm ; $\frac{R}{r} = e$ => most economic condition

$E_{max} = \frac{V}{r}$ -> $r = \frac{V}{E_{max}} = \frac{75}{60}$ = 1.25 cm

Overall diameter = $2R = 2 \times e \times r = 2 \times 2.718 \times 1.25 = 6.795$ cm.

Capacitance Grading :

In this method different insulating materials are placed between core and the outer most sheath. Due to this E_{max} and E_{min} difference is reduced so that cable can be operated safely at higher vtg compared to ungraded cable. This method is most practically used.

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$$E = \frac{q}{2\pi\epsilon_0\epsilon_r x}; \quad E_{max} = \frac{q}{2\pi\epsilon_0\epsilon_r r}; \quad E_{min} = \frac{q}{2\pi\epsilon_0\epsilon_r R}$$

$$x \cdot \epsilon_r = r\epsilon_r = R\epsilon_r = \text{const.} \Rightarrow E_{max} = E_{min}$$

x / ϵ_r / $x\epsilon_r$

1 / 40 (PVC) / 40

2 / 20 (RUB) / 40

4 / 10 (XLPE) / 40

Safe working vtg :

(1) Ungraded cable :

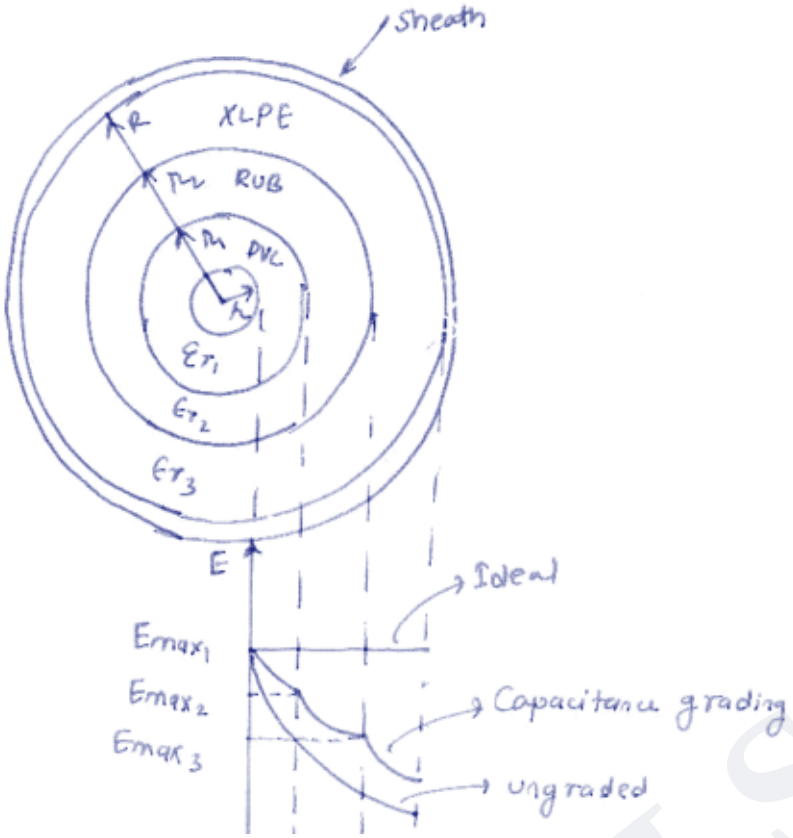
$$V_{max} = E_{max} r \ln\left(\frac{R}{r}\right)$$

(2) Capacitance Grading :

$$V_{max} = E_{max1} r \ln\left(\frac{r_1}{r}\right) + E_{max2} r_1 \ln\left(\frac{r_2}{r_1}\right) + E_{max3} r_2 \ln\left(\frac{R}{r_2}\right)$$

$$\frac{q}{2\pi\epsilon_0} = r\epsilon_{r1}E_{max1} = r_1\epsilon_{r2}E_{max2} = r_2\epsilon_{r3}E_{max3}$$

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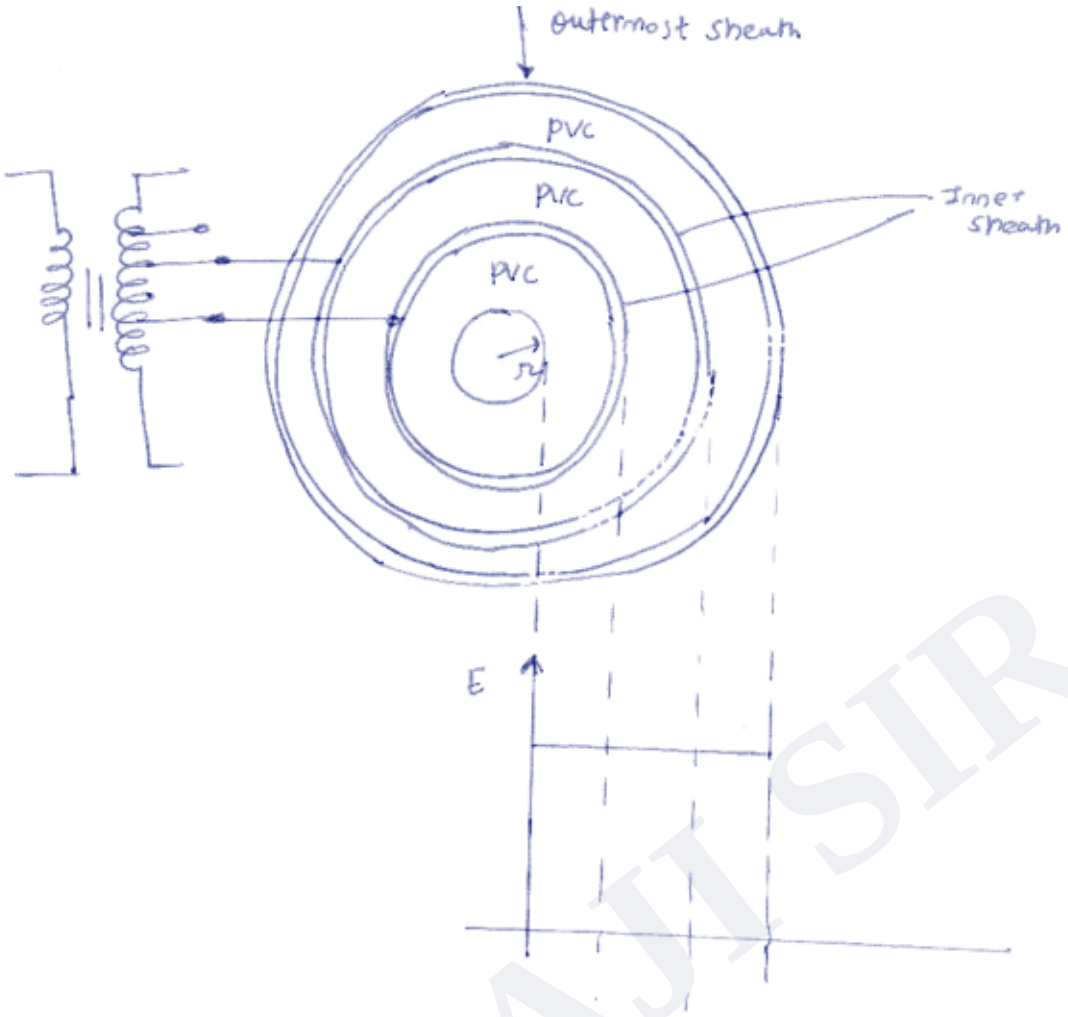
Three-layer graded cable: core radius r, dielectrics PVC (er1) to r1, RUB (er2) to r2, XLPE (er3) to R, then sheath; below, E against x: ungraded curve, capacitance grading steps, ideal flat line, with Emax1, Emax2, Emax3

Inner (or) Intersheath Grading :

* In this method between the core and the outermost sheath the same insulating material is used, but within this insulating material additional inner sheaths are placed; due to this cost is increased, and with additional X'mer maintenance cost is also increased.

* During transportation inner sheath may be damaged so that overall performance of cable is affected.

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Core radius r surrounded by three PVC layers separated by inner sheaths tapped from a transformer winding, outermost sheath; below, flat E against x

Eg: A single core lead sheathed cable is to be designed for 66 kV line to Earth. Its conductor radius is 10 mm and its insulating materials PVC, Rub, XLPE have relative permittivity of 5, 4 and 3 respectively and corresponding maximum permissible stresses are of 3.8, 2.6 and 2 kV/mm (rms) respectively. Find minimum diameter of lead sheath cable.

$$r\epsilon_{r1}E_{max1} = r_1\epsilon_{r2}E_{max2} = r_2\epsilon_{r3}E_{max3}$$

$$10 \times 5 \times 3.8 = r_1 \times 4 \times 2.6 = r_2 \times 3 \times 2 \rightarrow r_1 = 18.3 \text{ mm},\ r_2 = 31.7 \text{ mm}$$

$$66 = \left(3.8 \times 10 \times \ln\frac{18.3}{10}\right) + 2.6 \times 18.3 \ln\frac{31.7}{18.3} + 2 \times 31.7 \ln\frac{R}{31.7}$$

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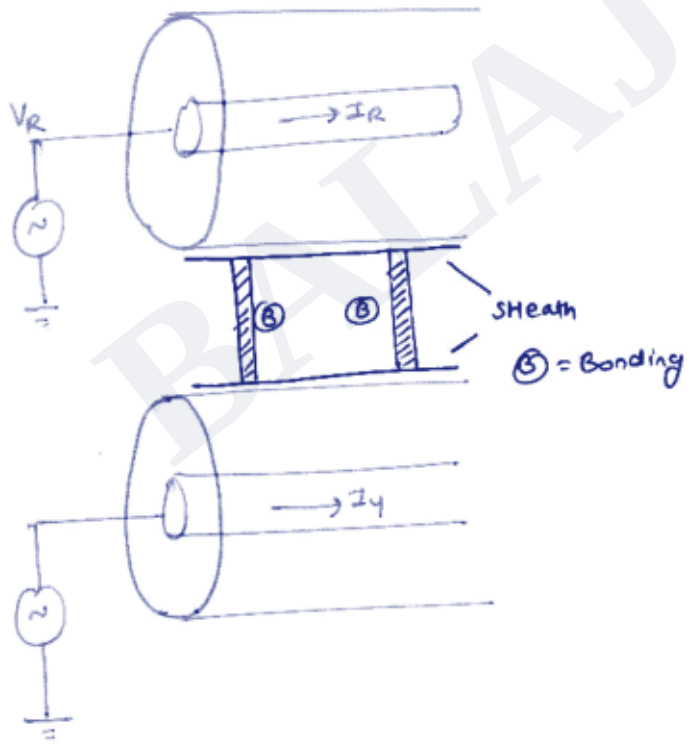
$R = 41.44\ mm \Rightarrow$ Diameter of cable $= 82.88\ mm$

$V_{max} = E_{max} r \ln\left(\frac{R}{r}\right) = 3.8 \times 10 \times \ln(41.44/10) = 52\ kV$ (ungraded, for comparison)

Bonding of the cable :

(1) In case of high capacity HVAC cable, due to electromagnetic induction, potential difference occurs between the sheaths of the adjacent phases. This will produce arcing within the phases of the cable, which will damage the cable due to overheating.
For protection of cable from this, metallic bonding is done between the phases by using bonding: internal resistance is increased, inductance is reduced, a leakage path is created for the current, and the cable can be safely operated at higher vtg.

Along with the bonding, surge vtg limiter can also be used for protection of the cable from lightning surges and over vtg.



Two single-core cables (phase R with VR, IR; phase Y with IY) whose sheaths are joined by metallic bonds (B)

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